

PINCHER CREEK, AB, Canada

WMO: 710850

Lat: 49.5222N

Lon: 114.0050W

Elev: 3903

StdP: 12.74

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.5	-12.9	-27.2	1.4	-19.3	-20.4	2.1	-12.7	48.2	33.8	43.4	33.4	6.9	280	0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	85.1	59.9	81.7	58.8	78.4	58.0	63.3	78.1	61.5	76.2	59.7	74.3	14.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	83.4	68.7	56.0	77.1	66.6	53.9	71.4	64.5	30.7	78.0	29.3	76.2	27.9	74.4	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.3	35.3	30.7	DB	-27.1	90.1	6.5	3.8	-31.8	92.9	-35.7	95.1	-39.3	97.2	-44.1	100.0	(4)
(5)			WB	-27.4	67.4	6.4	2.3	-32.1	69.0	-35.8	70.4	-39.5	71.7	-44.1	73.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	22.8	23.5	29.9	39.4	48.4	55.5	62.5	61.5	53.5	42.0	30.7	22.5		
	DBStd	18.21	15.76	14.66	13.91	9.42	7.88	5.85	5.49	5.89	8.36	9.79	13.29	14.81		
	HDD50	4451	843	742	624	332	125	16	1	3	58	276	580	851		
	HDD65	8790	1308	1161	1088	769	515	290	112	138	352	713	1028	1316		
	CDD50	1213	0	0	1	13	76	180	390	359	162	29	3	0		
	CDD65	74	0	0	0	0	1	4	35	28	6	0	0	0		
	CDH74	1824	0	0	0	2	46	109	756	706	201	4	0	0		
	CDH80	466	0	0	0	0	3	19	218	190	36	0	0	0		
	WSAvg	13.5	16.8	14.5	14.1	13.0	12.3	12.0	10.8	10.2	11.8	14.5	16.0	16.6		
	PrecAvg	27.4	2.0	1.7	2.0	2.0	3.1	3.8	1.9	1.8	2.0	2.0	2.5	2.1		
(15) Precipitation	PrecMax	36.4	3.1	2.8	3.9	3.7	6.0	10.2	6.6	5.3	6.0	3.5	5.5	4.6		
	PrecMin	20.1	0.6	0.5	0.7	1.1	0.6	1.5	0.3	0.4	0.2	0.8	0.6	0.9		
	PrecStd	5.0	0.7	0.7	0.8	0.6	1.5	2.0	1.5	1.1	1.3	0.9	1.4	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	52.6	59.8	70.4	79.0	83.1	90.0	89.2	84.6	70.8	59.7	49.4		
		MCWB	40.6	40.0	43.5	49.9	55.7	62.1	62.0	60.1	57.1	51.7	45.6	39.9		
	2%	DB	46.3	47.6	54.2	64.8	74.1	76.8	85.6	85.0	79.7	64.9	53.9	45.0		
		MCWB	37.0	38.0	41.8	47.6	53.6	58.1	61.1	58.9	55.8	49.2	43.6	36.9		
	5%	DB	43.2	43.9	50.2	59.5	69.2	72.9	81.9	81.5	75.3	60.3	49.9	41.9		
		MCWB	35.7	35.8	39.7	44.6	51.5	56.9	59.9	58.4	54.8	47.2	40.8	35.1		
	10%	DB	40.5	40.8	46.6	54.4	64.4	69.0	78.2	77.9	70.0	56.0	46.4	39.3		
		MCWB	34.3	33.8	37.5	42.0	49.2	54.5	59.6	57.6	52.7	44.8	38.6	33.3		
	0.4%	WB	41.9	41.4	45.6	51.1	57.6	64.2	66.9	64.1	59.7	53.7	47.9	41.1		
		MCDB	50.1	50.3	57.0	67.6	75.1	79.1	81.0	79.7	76.7	67.8	56.9	47.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.3	38.9	42.7	48.0	54.8	60.3	64.3	61.9	57.3	50.4	44.4	38.0		
		MCDB	44.6	46.2	52.7	62.8	71.3	73.6	78.8	77.7	75.1	62.8	53.2	43.6		
	5%	WB	36.3	36.4	40.3	45.3	52.5	58.0	62.5	60.2	55.5	47.8	41.4	35.6		
		MCDB	42.1	43.5	49.3	58.3	67.4	70.4	77.0	76.1	73.2	59.3	49.1	41.3		
	10%	WB	34.4	33.9	38.0	42.7	50.1	55.7	60.8	58.6	53.4	45.2	38.8	33.3		
		MCDB	40.5	40.5	46.1	53.7	62.9	66.5	75.0	74.5	68.8	54.9	46.0	39.2		
	5% DB	MDBR	16.2	17.2	17.8	19.7	21.3	20.9	26.0	26.8	23.3	18.3	14.9	14.7		
		MCDBR	15.2	18.1	21.5	28.5	30.7	29.2	33.6	34.8	33.8	24.1	17.1	13.6		
		MCWBR	10.5	11.7	12.8	15.2	15.0	14.9	14.4	14.2	14.9	12.8	10.6	9.8		
		MCWBR	14.9	18.5	20.7	26.7	28.3	26.7	29.4	30.2	31.0	23.2	17.5	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.254	0.282	0.339	0.344	0.337	0.336	0.350	0.315	0.282	0.259	0.232		
		taud	2.396	2.390	2.414	2.293	2.374	2.435	2.449	2.400	2.476	2.549	2.496	2.419		
	Ebn at Noon		266	290	298	287	290	292	291	281	282	276	259	256		
		Edh at Noon	20	25	30	38	36	35	34	34	28	22	18	16		
(44) All-Sky Solar Radiation	RadAvg		371	684	1046	1477	1760	1934	2194	1772	1245	759	404	292		
	RadStd		33	53	93	110	149	149	144	119	116	79	36	28		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page